IE 370 Airline Project

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3.1 Expected Rewards:

g(0,0) = 0

- Not selling any tickets so not making any money

g(1,0) = (¼ * 0) + (½ * 100) + (¼ * 100) = 75

- ¼ chance no one buys a ticket, ½ chance one person buys a ticket and, ¼ chance 2 people come to buy tickets so the one is sold.

g(0,1) = (¼ * 0) + (¾ * 0.4 * 200) = 60

- ¼ chance no one comes to buy a ticket, ½ chance one person come to buy a ticket and theres a 0.4 chance they buy the expensive ticket, also a ½ chance 2 people come to buy a ticket and 0.4 chance one of them buys the ticket so ¾ chance someone buys a ticket with chance of 0.4

g(2,0) = (¼ * 0) + (½ * 100) + (¼ * 200) = 100

- ¼ chance no one come to buy a ticket, ½ chance one person comes to buy a ticket and will automatically buy one cheap one, ¼ chance 2 people come to buy a ticket and both will automatically buy a $100 ticket

g(1,1) = (¼ * 0) + (½ * 100) + (¼ * (100 + (0.4 * 200)) = 95

- Given to us in the problem but also ¼ chance no one comes to buy a ticket, ½ chance one person comes to buy a ticket and then automatically buys the cheap ticket, ¼ chance 2 people come to buy a ticket, one automatically buys the cheap ticket the next then has a 0.4 chance of buying the expensive ticket

g(0,2) = (¼ * 0) + (½ * 0.4 * 200) + (¼ * 2 * 0.4 * 200) = 80

- ¼ chance no one comes to buy a ticket, ½ chance one person comes to buy a ticket and then buys an expensive ticket with chance of 0.4, ¼ chance 2 people come to buy tickets and each person will buy an expensive ticket with probability 0.4, since there's 2 of them multiply by 2

3.2 Transition Probabilities:

If k = (0,0):

- **Pk(i,i) = 1**, because our only actions are to sell tickets can't gain any and can't sell any if we have none to start

If k = (1,0)

- **Pk(i,i) = 0.25**, theres a ¼ chance nobody shows up to buy the cheap ticket

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- **Pk(i,i-1) = 0.75** because theres a ½ chance one person shows up and automatically buys the ticket and a ¼ chance 2 people show up and one of them will buy the ticket

If k = (0,1)

- **Pk(i,i) = ¼ + (¾ * 0.6) = 0.7**, ¼ chance nobody comes to buy a ticket, ½ chance one person come to buy and doesn't with probability 0.6 and ¼ chance 2 people come by and don't buy with probability 0.6
- **Pk(i,i-1) = ¾ * 0.4 = 0.3**, ½ chance one person comes and buys with chance of 0.4 or ¼ chance 2 people come buy and one buys with chance of 0.4

If k = (2,0)

- **Pk(i,i) = 0.25**, ¼ chance nobody comes to buy a ticket
- **Pk(i,i-1) = 0.5**, ½ chance only one person comes by and automatically buys a cheap ticket
- **Pk(i,i-2) = 0.25**, ¼ chance 2 people come by and automatically buy a cheap ticket

If k = (1,1)

- **Pk(i,i) = 0.25,** ¼ chance no one comes to buy a ticket
- **Pk(i,i-1) = ½ + (¼ * 0.6) = 0.65**, ½ chance one person comes and they will automatically buy the cheap ticket and then theres a ¼ chance 2 people come to buy a ticket, one will automatically buy the cheap one and the odds the second person doesnt buy will be 0.6
- **Pk(i,i-2) = ¼ * 0.4 = 0.1**, ¼ chance 2 people come buy, one will automatically buy the cheap ticket the next has probability 0.4 of buying the expensive ticket

If k = (0,2)

- **Pk(i,i) = ¼ + (½ * 0.6) + (¼ * 0.6^2) = 0.64**, ¼ chance nobody shows up to buy a ticket, ½ chance one person shows up to buy a ticket but wont buy with probability 0.6 and then ¼ chance 2 people show up to buy a ticket but both wont buy with probability 0.6 so 0.6^2
- **Pk(i,i-2) = ¼ * 0.4^2 = 0.04**, ¼ chance 2 people show up to buy a ticket and 0.4 chance for both of them that they buy so 0.4^2
- **Pk(i,i-1) 1 - 0.64 - 0.04 = 0.32**, All probabilities must add up to 1 so 0.32 is probability for this option

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3.3 Code To Solve MDP:

Initial Variable setup:

```
import numpy as np
#Variables
#Calculated rewards
g_actions = [0, 75, 60, 100, 95, 80]
#Calculated transition probability matrix
trans_prob = np.array([[1, 0, 0], [0.25, 0.75, 0], [0.7, 0.3, 0], [0.25, 0.5, 0.25], [0.25, 0.65, 0.1],[0.25, 0.32, 0.04]])
#Expected revenue empty matrix
Exp_Rev = np.zeros((11, 16))
#Optimal policy empty matrix
Opt_Policy = np.zeros((11, 16))
#If we have 0 tickets to sell on Day 10, then 0 tickets will be sold
Opt_Policy[10, 0] = 0
Exp_Rev[10, 0] = 0
#If we have 1 ticket to sell on Day 10, then the optimal decision is to sell 1 cheap ticket
Opt_Policy[10, 1] = 1
Exp_Rev[10, 1] = 75

#If we have more than 1 ticket on Day 10, then the optimal decision is to sell 2 cheaptickets
for tickets_remaining in range(2,16):
    Opt_Policy[10, tickets_remaining] = 3
    Exp_Rev[10, tickets_remaining] = 100
```

The code is just some initial variable setup, we have our the revenue of each action in the g_actions array. Then we have our transition probability matrix. The next 2 sections created our Expected Revenue and Optimal Policy matrices and just populated them with zero values. Next we already know some initial values for example if there's no tickets to sell on the last day then we cant sell any tickets or if theres 1 ticket to sell the last day we know the best option would be action , then if we have more than 1 ticket on the last day we know the optimal decision is action 3 so we populate our matrices with those already known values.

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Loop to solve MDP:

```
#Solve MDP
for D in range (1, 11):
    x = 10 - D
    for S in range (0, 16):
        options = np.zeros(6)
        if S == 0:
            options[0] = g_actions[0] + Exp_Rev[x + 1, S]
            Exp_Rev[x, S] = np.amax(options)
            #optimal decision
            Opt_Policy[x, S] = np.argmax(options)
            #optimal policy at state
        elif S == 1:
            options[0] = g_actions[0] + trans_prob[0,0] * Exp_Rev[x + 1, S] + trans_prob[0,1] *Exp_Rev[x + 1, S - 1]
            options[1] = g_actions[1] + trans_prob[1,0] * Exp_Rev[x + 1, S] + trans_prob[1,1] *Exp_Rev[x + 1, S - 1]
            options[3] = g_actions[3] + trans_prob[3,0] * Exp_Rev[x + 1, S] + trans_prob[3,1] *Exp_Rev[x + 1, S - 1]
            Exp_Rev[x, S] = np.amax(options)
            #Max potential revenue at state
            Opt_Policy[x, S] = np.argmax(options)
            #optimal policy at state
        else:
            for A in range (0, 6):
                options[A] = g_actions[A] + trans_prob[A,0] * Exp_Rev[x + 1, S] + trans_prob[A,1] *Exp_Rev[x + 1, S - 1] + tra
                Exp_Rev[x, S] = np.amax(options)
                #Max revenue at state
                Opt_Policy[x, S] = np.argmax(options)
                #optimal policy at state
print(Opt_Policy)
print(Exp_Rev, 2)
```

This is a for loop that iterates over all the days and amount of tickets available to sell for each day and calculates the expected revenue for each action and then selects the max value of that and populates our optimal policy/expected revenue arrays. These calculations use dynamic programming that calculates revenues based on expected revenue of future states and transition probabilities to get in between states.

Print Results:

```
#Print Out Results In Organized Fashion
for row_index, row in enumerate(Opt_Policy):
    print(f"Time: {row_index}")
    for col_index, value in enumerate(row):
        print(f"If In State {col_index}: Optimal Policy is {Opt_Policy[row_index][col_index]} with value {Exp_Rev[row_index][c
    print("\n")
```

The above code is just a double for loop that walks through the Optimal Policy and Expected Revenue matrices for each time and prints out the optimal policy at each state

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Results:

Note In these Results Optimal Policy Corresponds to the Indexs of this array [(0,0), (1,0), (0,1), (2,0), (1,1), (0,2)] i.e. Policy 0.0 would be action (0,0), Policy 3.0 would be (2,0), Policy 2.0 would be (0,1) etc.

```
Time: 0
If In State 0: Optimal Policy is 0.0 with value 0.0
If In State 1: Optimal Policy is 3.0 with value 133.33327770233154
If In State 2: Optimal Policy is 2.0 with value 326.3355425401123
If In State 3: Optimal Policy is 2.0 with value 494.2741129440917
If In State 4: Optimal Policy is 2.0 with value 622.3187059780272
If In State 5: Optimal Policy is 2.0 with value 713.8551502373045
If In State 6: Optimal Policy is 2.0 with value 784.7736005374511
If In State 7: Optimal Policy is 4.0 with value 848.6753989881714
If In State 8: Optimal Policy is 4.0 with value 910.1556734081179
If In State 9: Optimal Policy is 4.0 with value 964.9077163701418
If In State 10: Optimal Policy is 4.0 with value 1009.0233542763185
If In State 11: Optimal Policy is 4.0 with value 1041.8022715326906
If In State 12: Optimal Policy is 3.0 with value 1066.5918354537353
If In State 13: Optimal Policy is 3.0 with value 1083.032731678467
If In State 14: Optimal Policy is 3.0 with value 1092.5143684844973
If In State 15: Optimal Policy is 3.0 with value 1097.1881174011232


Time: 1
If In State 0: Optimal Policy is 0.0 with value 0.0
If In State 1: Optimal Policy is 3.0 with value 133.33311080932617
If In State 2: Optimal Policy is 2.0 with value 323.3365847104492
If In State 3: Optimal Policy is 2.0 with value 481.8187679013671
If In State 4: Optimal Policy is 2.0 with value 596.818679439453
If In State 5: Optimal Policy is 2.0 with value 678.2993520078124
If In State 6: Optimal Policy is 2.0 with value 744.691135621582
If In State 7: Optimal Policy is 4.0 with value 807.1849025334474
If In State 8: Optimal Policy is 4.0 with value 864.0654927968752
If In State 9: Optimal Policy is 4.0 with value 910.1866231953127
If In State 10: Optimal Policy is 4.0 with value 943.981999678711
If In State 11: Optimal Policy is 3.0 with value 968.7812376879883
If In State 12: Optimal Policy is 3.0 with value 984.822866760254
If In State 13: Optimal Policy is 3.0 with value 993.7039555053711
If In State 14: Optimal Policy is 3.0 with value 997.8266961669922
If In State 15: Optimal Policy is 3.0 with value 999.3951217651368
```

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```
Time: 2
If In State 0: Optimal Policy is 0.0 with value 0.0
If In State 1: Optimal Policy is 3.0 with value 133.3324432373047
If In State 2: Optimal Policy is 2.0 with value 319.05264534179685
If In State 3: Optimal Policy is 2.0 with value 465.86139185546864
If In State 4: Optimal Policy is 2.0 with value 567.2289455468749
If In State 5: Optimal Policy is 2.0 with value 640.1866690625
If In State 6: Optimal Policy is 4.0 with value 703.7644784326172
If In State 7: Optimal Policy is 4.0 with value 762.8772985839846
If In State 8: Optimal Policy is 4.0 with value 811.2752034960938
If In State 9: Optimal Policy is 4.0 with value 846.2800442578125
If In State 10: Optimal Policy is 3.0 with value 871.0898022460938
If In State 11: Optimal Policy is 3.0 with value 886.6653020019531
If In State 12: Optimal Policy is 3.0 with value 894.8710607910157
If In State 13: Optimal Policy is 3.0 with value 898.4083984375
If In State 14: Optimal Policy is 3.0 with value 899.6189270019531
If In State 15: Optimal Policy is 3.0 with value 899.9342346191406


Time: 3
If In State 0: Optimal Policy is 0.0 with value 0.0
If In State 1: Optimal Policy is 3.0 with value 133.32977294921875
If In State 2: Optimal Policy is 2.0 with value 312.93387636718745
If In State 3: Optimal Policy is 2.0 with value 445.68746992187494
If In State 4: Optimal Policy is 2.0 with value 533.603863671875
If In State 5: Optimal Policy is 2.0 with value 600.150728515625
If In State 6: Optimal Policy is 4.0 with value 661.2244741210939
If In State 7: Optimal Policy is 4.0 with value 712.2652702148438
If In State 8: Optimal Policy is 4.0 with value 748.7213217773439
If In State 9: Optimal Policy is 3.0 with value 773.5386323242187
If In State 10: Optimal Policy is 3.0 with value 788.5606225585938
If In State 11: Optimal Policy is 3.0 with value 796.0013305664063
If In State 12: Optimal Policy is 3.0 with value 798.9209594726562
If In State 13: Optimal Policy is 3.0 with value 799.7903442382812
If In State 14: Optimal Policy is 3.0 with value 799.9740600585938
If In State 15: Optimal Policy is 3.0 with value 799.9984741210938
```

```
Time: 4
If In State 0: Optimal Policy is 0.0 with value 0.0
If In State 1: Optimal Policy is 3.0 with value 133.319091796875
If In State 2: Optimal Policy is 2.0 with value 304.19735546874995
If In State 3: Optimal Policy is 2.0 with value 420.61180468749995
If In State 4: Optimal Policy is 2.0 with value 496.31474609375005
If In State 5: Optimal Policy is 4.0 with value 558.9375781250001
If In State 6: Optimal Policy is 4.0 with value 613.134294921875
If In State 7: Optimal Policy is 4.0 with value 651.3368828125001
If In State 8: Optimal Policy is 3.0 with value 676.155673828125
If In State 9: Optimal Policy is 3.0 with value 690.506298828125
If In State 10: Optimal Policy is 3.0 with value 697.07421875
If In State 11: Optimal Policy is 3.0 with value 699.3505859375
If In State 12: Optimal Policy is 3.0 with value 699.908447265625
If In State 13: Optimal Policy is 3.0 with value 699.993896484375
If In State 14: Optimal Policy is 3.0 with value 700.0
If In State 15: Optimal Policy is 3.0 with value 700.0


Time: 5
If In State 0: Optimal Policy is 0.0 with value 0.0
If In State 1: Optimal Policy is 3.0 with value 133.2763671875
If In State 2: Optimal Policy is 2.0 with value 291.73492187499994
If In State 3: Optimal Policy is 2.0 with value 390.13046875
If In State 4: Optimal Policy is 4.0 with value 456.10800781250003
If In State 5: Optimal Policy is 4.0 with value 513.8173046875
If In State 6: Optimal Policy is 4.0 with value 554.168984375
If In State 7: Optimal Policy is 3.0 with value 578.98125
If In State 8: Optimal Policy is 3.0 with value 592.4912109375
If In State 9: Optimal Policy is 3.0 with value 598.0615234375
If In State 10: Optimal Policy is 3.0 with value 599.6826171875
If In State 11: Optimal Policy is 3.0 with value 599.9755859375
If In State 12: Optimal Policy is 3.0 with value 600.0
If In State 13: Optimal Policy is 3.0 with value 600.0
If In State 14: Optimal Policy is 3.0 with value 600.0
If In State 15: Optimal Policy is 3.0 with value 600.0
```

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```
Time: 6
If In State 0: Optimal Policy is 0.0 with value 0.0
If In State 1: Optimal Policy is 3.0 with value 133.10546875
If In State 2: Optimal Policy is 2.0 with value 274.0046875
If In State 3: Optimal Policy is 2.0 with value 354.184375
If In State 4: Optimal Policy is 4.0 with value 413.95078125000003
If In State 5: Optimal Policy is 4.0 with value 457.3234375
If In State 6: Optimal Policy is 3.0 with value 482.0546875
If In State 7: Optimal Policy is 3.0 with value 494.4921875
If In State 8: Optimal Policy is 3.0 with value 498.92578125
If In State 9: Optimal Policy is 3.0 with value 499.90234375
If In State 10: Optimal Policy is 3.0 with value 500.0
If In State 11: Optimal Policy is 3.0 with value 500.0
If In State 12: Optimal Policy is 3.0 with value 500.0
If In State 13: Optimal Policy is 3.0 with value 500.0
If In State 14: Optimal Policy is 3.0 with value 500.0
If In State 15: Optimal Policy is 3.0 with value 500.0


Time: 7
If In State 0: Optimal Policy is 0.0 with value 0.0
If In State 1: Optimal Policy is 3.0 with value 132.421875
If In State 2: Optimal Policy is 2.0 with value 248.96875
If In State 3: Optimal Policy is 4.0 with value 313.5625
If In State 4: Optimal Policy is 4.0 with value 360.953125
If In State 5: Optimal Policy is 3.0 with value 385.390625
If In State 6: Optimal Policy is 3.0 with value 396.484375
If In State 7: Optimal Policy is 3.0 with value 399.609375
If In State 8: Optimal Policy is 3.0 with value 400.0
If In State 9: Optimal Policy is 3.0 with value 400.0
If In State 10: Optimal Policy is 3.0 with value 400.0
If In State 11: Optimal Policy is 3.0 with value 400.0
If In State 12: Optimal Policy is 3.0 with value 400.0
If In State 13: Optimal Policy is 3.0 with value 400.0
If In State 14: Optimal Policy is 3.0 with value 400.0
If In State 15: Optimal Policy is 3.0 with value 400.0
```

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```
Time: 8
If In State 0: Optimal Policy is 0.0 with value 0.0
If In State 1: Optimal Policy is 3.0 with value 129.6875
If In State 2: Optimal Policy is 4.0 with value 214.375
If In State 3: Optimal Policy is 4.0 with value 265.0
If In State 4: Optimal Policy is 3.0 with value 289.0625
If In State 5: Optimal Policy is 3.0 with value 298.4375
If In State 6: Optimal Policy is 3.0 with value 300.0
If In State 7: Optimal Policy is 3.0 with value 300.0
If In State 8: Optimal Policy is 3.0 with value 300.0
If In State 9: Optimal Policy is 3.0 with value 300.0
If In State 10: Optimal Policy is 3.0 with value 300.0
If In State 11: Optimal Policy is 3.0 with value 300.0
If In State 12: Optimal Policy is 3.0 with value 300.0
If In State 13: Optimal Policy is 3.0 with value 300.0
If In State 14: Optimal Policy is 3.0 with value 300.0
If In State 15: Optimal Policy is 3.0 with value 300.0


Time: 9
If In State 0: Optimal Policy is 0.0 with value 0.0
If In State 1: Optimal Policy is 3.0 with value 118.75
If In State 2: Optimal Policy is 4.0 with value 168.75
If In State 3: Optimal Policy is 3.0 with value 193.75
If In State 4: Optimal Policy is 3.0 with value 200.0
If In State 5: Optimal Policy is 3.0 with value 200.0
If In State 6: Optimal Policy is 3.0 with value 200.0
If In State 7: Optimal Policy is 3.0 with value 200.0
If In State 8: Optimal Policy is 3.0 with value 200.0
If In State 9: Optimal Policy is 3.0 with value 200.0
If In State 10: Optimal Policy is 3.0 with value 200.0
If In State 11: Optimal Policy is 3.0 with value 200.0
If In State 12: Optimal Policy is 3.0 with value 200.0
If In State 13: Optimal Policy is 3.0 with value 200.0
If In State 14: Optimal Policy is 3.0 with value 200.0
If In State 15: Optimal Policy is 3.0 with value 200.0
```

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```
Time: 10
If In State 0: Optimal Policy is 0.0 with value 0.0
If In State 1: Optimal Policy is 1.0 with value 75.0
If In State 2: Optimal Policy is 3.0 with value 100.0
If In State 3: Optimal Policy is 3.0 with value 100.0
If In State 4: Optimal Policy is 3.0 with value 100.0
If In State 5: Optimal Policy is 3.0 with value 100.0
If In State 6: Optimal Policy is 3.0 with value 100.0
If In State 7: Optimal Policy is 3.0 with value 100.0
If In State 8: Optimal Policy is 3.0 with value 100.0
If In State 9: Optimal Policy is 3.0 with value 100.0
If In State 10: Optimal Policy is 3.0 with value 100.0
If In State 11: Optimal Policy is 3.0 with value 100.0
If In State 12: Optimal Policy is 3.0 with value 100.0
If In State 13: Optimal Policy is 3.0 with value 100.0
If In State 14: Optimal Policy is 3.0 with value 100.0
If In State 11: Optimal Policy is 3.0 with value 100.0
If In State 12: Optimal Policy is 3.0 with value 100.0
If In State 13: Optimal Policy is 3.0 with value 100.0
If In State 14: Optimal Policy is 3.0 with value 100.0
If In State 13: Optimal Policy is 3.0 with value 100.0
If In State 14: Optimal Policy is 3.0 with value 100.0
If In State 15: Optimal Policy is 3.0 with value 100.0
```